OR4023 (Optimization) Quiz 2

Name:_____ Student ID:_____

Instructions:

1. No calculator is allowed.
2. Duration of quiz: 90 minutes

Question 1. (20 marks) Consider the problem

$$\text{minimize} f(x)=x^4-1$$

Solve this problem using Newton's method. Start from $x_0=4$ and perform 2 iterations. Prove that the iterates converge to the solution. What is the order of convergence?

Answer. We have $f'(x)=4x^3, f''(x)=12x^2$, so $f'(x)/f''(x)=x/3$.

- iteration 1:

$$x_1=x_0-\frac{f'(x_0)}{f''(x_0)}=4-\frac{4}{3}=\frac{8}{3}.$$

- iteration 2:

$$x_2=x_1-\frac{f'(x_1)}{f''(x_1)}=\frac{8}{3}-\frac{8}{9}=\frac{16}{9}.$$

In general, we have

$$x_{n+1}=x_n-\frac{f'(x_n)}{f''(x_n)}=\frac{2}{3}x_n.$$

Hence

$$x_n=4\left(\frac{2}{3}\right)^n\to 0$$

as $n\to\infty$. It's easy to see 0 is the solution. The order of convergence is 1, since

$$\lim_{n\to\infty}\left|\frac{x_{n+1}-0}{x_n-0}\right|=\frac{2}{3}.$$

Question 2. (20 marks) Consider the problem

$$\text{minimize } f(x_1,x_2)=(x_1-2x_2)^2+x_1^4.$$

a) Suppose a Newton's method with a line search is used to minimize the function, starting from the point $x=(2,1)^T$. What is the Newton search direction at this point?

b) Does the trial step size $\alpha=1$ satisfy the sufficient decrease condition for $\mu=0.2$?

Answer. We have

$$\nabla f(x_1,x_2)=\begin{pmatrix} 2(x_1-2x_2)+4x_1^3 \\ -4(x_1-2x_2)\end{pmatrix},\quad \nabla^2 f(x_1,x_2)=\begin{pmatrix} 2+12x_1^2 & -4 \\ -4 & 8\end{pmatrix}.$$

a) The Newton search direction at $(2,1)^T$ is

$$p=-[\nabla^2 f(2,1)]^{-1}\nabla f(2,1)=-\begin{pmatrix} 50 & -4 \\ -4 & 8\end{pmatrix}^{-1}\begin{pmatrix} 32 \\ 0\end{pmatrix}=\begin{pmatrix} -\frac{2}{3} \\ -\frac{1}{3}\end{pmatrix}.$$

b) We have $x+\alpha p=\left(\frac{4}{3},\frac{2}{3}\right)$, and

$$f(x+\alpha p)-f(x)=\left(\frac{4}{3}\right)^4-2^4=-\frac{1040}{81},$$

$$\mu\alpha p^T\nabla f(x)=-\frac{64}{15}.$$

Since

$$f(x+\alpha p)-f(x)<\mu\alpha p^T\nabla f(x),$$

The sufficient decrease condition is satisfied.

Question 3. (10 marks) Show that the BFGS update formula

$$B_{k+1}=B_k-\frac{(B_ks_k)(B_ks_k)^T}{s_k^TB_ks_k}+\frac{y_ky_k^T}{y_k^Ts_k}$$

meets the secant condition and preserves symmetry.

Answer. We have

$$\begin{aligned}
B_{k+1}s_k &= B_ks_k-\frac{(B_ks_k)(B_ks_k)^Ts_k}{s_k^TB_ks_k}+\frac{y_ky_k^Ts_k}{y_k^Ts_k}\\
&= B_ks_k-\frac{(B_ks_k)s_k^TB_k{}^Ts_k}{s_k^TB_ks_k}+y_k\\
&= B_ks_k-\frac{(B_ks_k)s_k^TB_ks_k}{s_k^TB_ks_k}+y_k\\
&= B_ks_k-B_ks_k+y_k\\
&= y_k.
\end{aligned}$$

Hence the secant condition is met. In addition,

$$B_{k+1}^T=B_k^T-\frac{(B_ks_k)(B_ks_k)^T}{s_k^TB_ks_k}+\frac{y_ky_k^T}{y_k^Ts_k}=B_{k+1}.$$

Hence symmetry is preserved.

Question 4. (25 marks) Use the method of Lagrangian multipliers to solve

$$\begin{array}{ll}\text{minimize} & f(\mathbf{x})=x_1^2+2x_1x_2+x_2^2\\ \text{subject to} & 3x_1^2+x_2^2=9\end{array}$$

Answer. Let $h(x)=3x_1^2+x_2^2-9$. Define

$$L(x,\lambda)=(x_1^2+2x_1x_2+x_2^2)+\lambda(3x_1^2+x_2^2-9).$$

Then

$$\nabla_xL(x,\lambda)=\begin{pmatrix}2x_1+2x_2+6\lambda x_1\\ 2x_1+2x_2+2\lambda x_2\end{pmatrix}=0\Longrightarrow\lambda(3x_1-x_2)=0.$$

$$\nabla_{xx}L(x,\lambda)=\begin{pmatrix}2+6\lambda & 2\\ 2 & 2+2\lambda\end{pmatrix},\quad \nabla h(x)=\begin{pmatrix}6x_1\\ 2x_2\end{pmatrix}.$$

- case 1, $\lambda=0$. Then $2x_1+2x_2=0\Rightarrow x_2=-x_1$. Plugging into the constraint $h(x)=0$, we have

$$x_1=\pm\frac{3}{2},\quad x_2=\mp\frac{3}{2},\quad \nabla_{xx}L(x,\lambda)=\begin{pmatrix}2 & 2\\ 2 & 2\end{pmatrix}$$

  - $\mathbf{x}=\left(\frac{3}{2},-\frac{3}{2}\right)^T$, then $\nabla h(x)=(9,-3)^T$. If $y^T\nabla h(x)=0$, then $y_2=3y_1$. For those y, with $y\neq0$, we have

$$y^T\begin{pmatrix}2 & 2\\ 2 & 2\end{pmatrix}y=2y_1^2+4y_1y_2+2y_2^2=32y_1^2>0.$$

    Hence the 2nd order sufficient condition is satisfied. Thus $\mathbf{x}=\left(\frac{3}{2},-\frac{3}{2}\right)^T$ is a **local min**.

- $\mathbf{x}=\left(-\frac{3}{2},\frac{3}{2}\right)^T$, then $\nabla h(x)=(-9,3)^T$. If $y^T\nabla h(x)=0$, then $y_2=3y_1$. Similar to the previous case, we know $\mathbf{x}=\left(-\frac{3}{2},\frac{3}{2}\right)^T$ is a **local min**.

- case 2, $3x_1-x_2=0$. Plugging into the constraint $h(x)=0$, we obtain

$$x_1=\pm\frac{\sqrt{3}}{2},\quad x_2=\pm\frac{3\sqrt{3}}{2}.$$

From $2x_1+2x_2+6\lambda x_1=0$, we obtain $\lambda=-\frac{4}{3}$. Then

$$\nabla_{xx}L(x,\lambda)=\begin{pmatrix}-6 & 2\\ 2 & -\frac{2}{3}\end{pmatrix}.$$

  - $\mathbf{x}=\left(\frac{\sqrt{3}}{2},\frac{3\sqrt{3}}{2}\right)$, then $\nabla h(x)=(3\sqrt{3},3\sqrt{3})^T$. If $y^T\nabla h(x)=0$, then $y_2=-y_1$. For those y, with $y\neq 0$, we have

$$y^T\begin{pmatrix}-6 & 2\\ 2 & -\frac{2}{3}\end{pmatrix}y=-6y_1^2+4y_1y_2+-\frac{2}{3}y_2^2=-\frac{32}{3}y_1^2<0$$

    Thus it meets the 2nd order sufficient condition for **local max**.
  - $\mathbf{x}=\left(-\frac{\sqrt{3}}{2},-\frac{3\sqrt{3}}{2}\right)$, also a **local max**.

Question 5. (25 marks) Suppose the logarithmic barrier method is used to solve the problem

$$\begin{array}{ll}\text{minimize} & f(\mathbf{x})=\frac{1}{2}x_1^2+\frac{1}{2}x_2^2\\ \text{subject to} & x_1+x_2\geq 1.\end{array}$$

What is $\mathbf{x}(\mu)$? What is $\lambda(\mu)$? What is $\mathbf{x}^*$? What is λ^* ?

Answer. Let $g(x)=x_1+x_2-1$. The barrier function is

$$\beta(x,\mu)=\left(\frac{1}{2}x_1^2+\frac{1}{2}x_2^2\right)-\mu\log(x_1+x_2-1),\quad \mu>0.$$

The 1st order necessary condition yields

$$\begin{pmatrix}x_1-\frac{\mu}{x_1+x_2-1}\\ x_2-\frac{\mu}{x_1+x_2-1}\end{pmatrix}=0\Longrightarrow x_1=x_2\Longrightarrow x_1-\frac{\mu}{2x_1-1}\Longrightarrow 2x_1^2-x_1-\mu=0.$$

Hence

$$x_1=x_2=\frac{1+\sqrt{1+8\mu}}{4}.$$

The "minus" solution is discarded due to the constraint $x_1+x_2\geqslant 1$. Thus

$$\mathbf{x}(\mu)=\left(\frac{1+\sqrt{1+8\mu}}{4},\frac{1+\sqrt{1+8\mu}}{4}\right)^T,$$

and

$$g(\mathbf{x}(\mu))=\frac{1+\sqrt{1+8\mu}}{2}-1=\frac{-1+\sqrt{1+8\mu}}{2}.$$

$$\lambda(\mu)=\frac{\mu}{g(\mathbf{x})}=\frac{2\mu}{-1+\sqrt{1+8\mu}}.$$

As $\mu\to 0$, we have

$$\mathbf{x}(\mu)\to\mathbf{x}^*=\left(\frac{1}{2},\frac{1}{2}\right)^T,\quad \lambda(\mu)=\frac{2\mu(1+\sqrt{1+8\mu})}{(-1+\sqrt{1+8\mu})(1+\sqrt{1+8\mu})}=\frac{2\mu(1+\sqrt{1+8\mu})}{8\mu}\to\frac{1}{2}.$$